Brain–Model Alignment Across Neural Recording Modalities

NX-414 Brain-like Computation and Intelligence — Project Report

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1 Datasets and Exploratory Analysis

We use three datasets: **TVSD** (macaque ephys, V1/V4/IT; 22,248 train / 100 test, 2 subjects), **THINGS-EEG2** (human EEG, stimuli × channels × 80 timepoints at 100 Hz; 16,540 / 200, 10 subjects), and **NSD** (human fMRI, 28 ROIs; ∼9k / ∼1k, 8 subjects). Two models are compared: **Model A** (ResNet-152, adversarial) and **Model B** (Qwen3-VL-2B), each with 10 layers projected to 30k dims.

EEG noise ceilings. The **variance-based** [1] method matches stored ceilings exactly ($r = 1.000$, MAE $= 0.00\%$), while **split-half** [2] underestimates them ($r = 0.931$, MAE $= 4.84\%$, paired $t = -53.5$, $p \approx 0$). Both recover the same temporal structure, with reliability peaking at 100–400 ms in posterior channels (Figure 1).

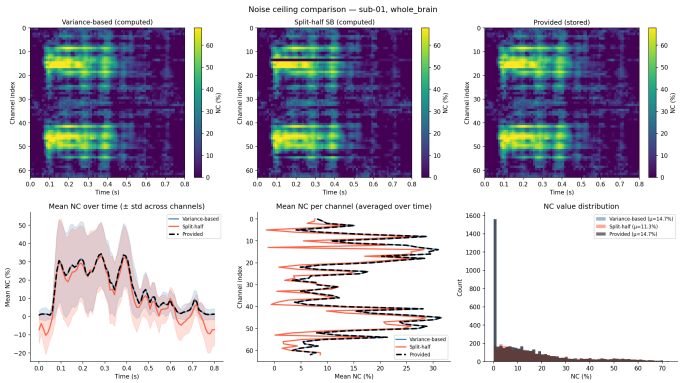

Figure 1: EEG noise ceiling comparison (subject 01): variance-based vs. split-half vs. stored.

NSD reliability. Noise ceilings peak in occipital and fusiform cortex (NC > 0.29) and fall to noise floor in motor/frontal areas (NC < 0.03), confirming that alignment analyses should target visual ROIs (Figure 2).

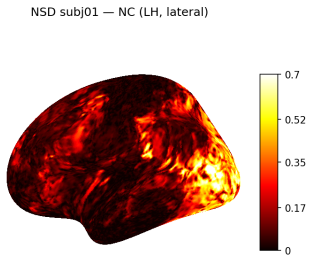

Figure 2: NSD noise ceiling on fsaverage LH (subject 01).

2 Brain-Model Alignment

2.1 Representational Alignment (RSA & CKA)

We compared model and neural representations using RSA and unbiased linear CKA on test stimuli only, without noise-ceiling correction. ResNet152-adv outperforms Qwen3-VL-2B on TVSD across ROIs (e.g. IT: RSA $= 0.22$ vs. 0.11; CKA $= 0.21$ vs. 0.14), with both metrics agreeing on ROI rankings and layer trends. EEG alignment peaks around 100 ms then declines. RSA and CKA diverge for ResNet152-adv on TVSD V1 (CKA $= 0.33$ vs. RSA $= 0.17$), suggesting low-dimensional alignment inflates CKA (Figure 3).

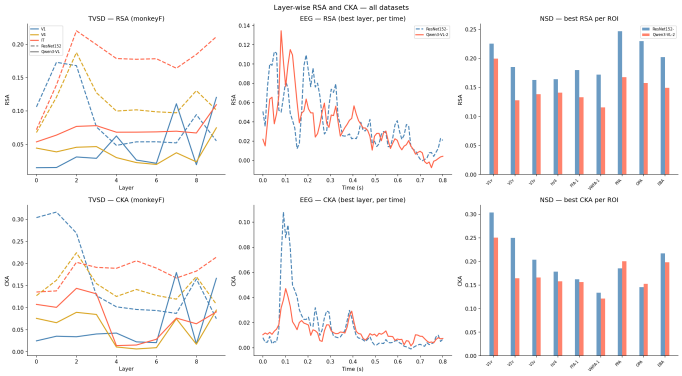

Figure 3: Layer-wise RSA (top) and CKA (bottom).

2.2 Linear Encoding Models

We fit ridge regression from PCA-compressed (512D) model features to neural responses (5-fold CV regularisation), reporting Pearson r, explained variance (EV), and noise-ceiling-corrected versions (r_{NC}, $\mathrm{EV}_{\mathrm{NC}}$). ResNet152-adv outperforms Qwen3-VL-2B on all TVSD ROIs (e.g. monkeyF IT: $r_{\mathrm{NC}} = 0.768$ vs. 0.713), with best layers progressing from V1 to IT, matching the ventral hierarchy. On NSD, Qwen3-VL-2B performs better across ROIs (e.g. V1v: 0.835 vs. 0.821), especially in higher-level areas. EEG performance is similar for both ($r_{\mathrm{NC}} \approx 0.43$).

Table 1: Best-layer r_{NC} per target.

Dataset	Target	Qwen	ResNet	Δ
EEG	occ. parietal	0.430	0.418	+0.012
NSD	V1v	0.835	0.821	+0.014
	PPA	0.824	0.793	+0.031
	EBA	0.796	0.745	+0.051
TVSD	mF/V1	0.797	0.806	−0.008
	mF/IT	0.713	0.768	−0.054

2.3 Encoding-RSA and Encoding-CKA

Encoding-RSA and Encoding-CKA test whether linear models preserve neural geometry. For TVSD and NSD they closely track predictive metrics—layers with high r_{NC} also show high Encoding-RSA/CKA—indicating that good readouts recover population geometry. For EEG, metric disagreement shows that geometric similarity and linear predictability can diverge (Figure 4).

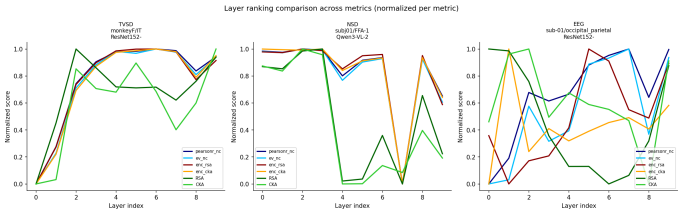

Figure 4: Layer ranking across all six metrics (normalised) for three representative cases.

3 Open-ended Extension

3.1 Motivation

The baseline pipeline selects a single model layer per ROI and compresses activations to 512 PCA dimensions. This is restrictive: early layers encode edges and textures while late layers encode object identity, so a multi-layer readout could explain variance no single layer captures. We ask whether combining features from multiple ResNet-152 layers improves alignment over the best single layer, and whether the benefit differs across the ventral hierarchy. Our hypothesis is that gains will be largest in V4 and IT, where no single layer dominates, and smallest in V1.

3.2 Methods

We focus on TVSD monkeyF (V1, V4, IT) with ResNet152-adv. Each of the 10 projected layers is independently compressed to $k = 128$ PCA dimensions (fit on training stimuli). We evaluate six fixed layer subsets (Table 2) plus an exhaustive sweep over all $\binom{10}{2} = 45$ two-layer pairs. For each subset, compressed representations are concatenated and a RidgeCV model (5-fold CV, $\alpha \in [10^{-2}, 10^{4}]$) is fit on training and evaluated on test stimuli.

Table 2: Layer subsets (indices 0–9: earliest to latest).

Subset	Indices	n
Best-1 (baseline)	best per ROI	1
Early-2	0, 1	2
Mid-2	4, 5	2
Late-2	8, 9	2
Consec-3	3, 4, 5	3
Skip-3	0, 4, 9	3
All-10	0–9	10

3.3 Results

Multi-layer readouts consistently outperform the single-layer baseline, but gains are strongly ROI-dependent (Figures 5–6). For **V1**, the baseline (layer 1) already captures most variance; only Early-2 and All-10 give modest gains (+0.015, +0.014), while mid/late subsets hurt (Mid-2: −0.122). For **V4**, late-layer content is essential: Early-2 drops −0.085, while Skip-3 (+0.012) and All-10 (+0.052) improve. For **IT**, any combination including late layers helps—Late-2 (+0.024), Skip-3 (+0.034), All-10 (+0.049)—confirming that IT integrates information across full network depth. Crucially, Skip-3 $[0, 4, 9]$ recovers most of the All-10 gain at one-third the dimensionality (mean $r_{\mathrm{NC}} = 0.748$ vs. 0.773 for All-10 and 0.735 for Best-1), suggesting *hierarchical diversity* matters more than the number of layers. The pairwise heatmap confirms this: the largest Δr_{NC} values concentrate off-diagonal (one early + one late layer), while adjacent pairs rarely beat the baseline.

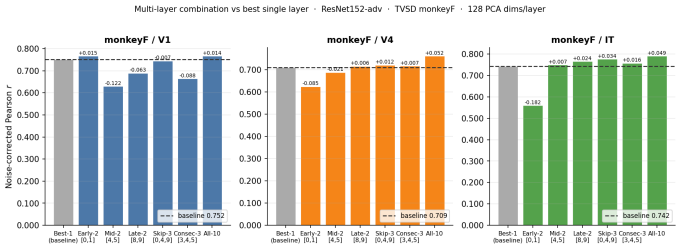

Figure 5: r_{NC} per layer subset across V1, V4, IT. Dashed line: single-best-layer baseline.

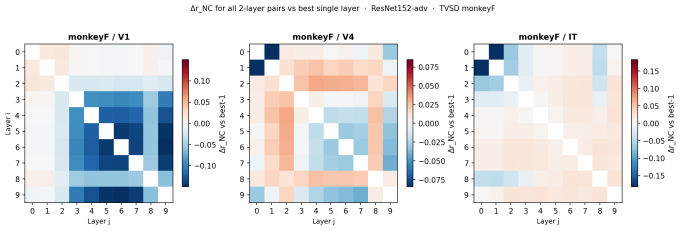

Figure 6: Δr_{NC} for all 45 two-layer pairs (monkeyF). Blue = gain; red = loss.

3.4 Scientific Insights

The results reveal a clear gradient along the ventral stream: V1 is near-optimally described by the earliest ResNet layer, while IT requires contributions across the full network depth, consistent with increasing receptive field size, selectivity complexity, and invariance from V1 to IT. A limitation is the fixed $k = 128$ compression, which may not reflect each layer's true intrinsic dimensionality. Additionally, results are limited to monkeyF and ResNet152-adv; generalisation to monkeyN or Qwen3-VL-2B remains open.

4 Conclusion

EEG was the noisiest modality, with noise ceilings below 20% even in reliable channels; NSD fMRI and TVSD ephys were more tractable, with V1/V1v yielding the highest noise-corrected scores. ResNet152-adv was the stronger predictor for macaque electrophysiology; Qwen3-VL-2B held a consistent edge in human fMRI, especially in semantic ROIs (FFA-1, PPA). RSA and CKA agreed on layer and model rankings but diverged in absolute magnitude (notably TVSD V1 where CKA substantially exceeded RSA for ResNet152-adv). The multi-layer extension showed that hierarchically diverse feature combinations outperform single-layer baselines, with the largest gains in higher visual areas. Key limitations

include single-subject analyses, fixed PCA compression, and collapsed EEG time axes; natural extensions include multi-subject averaging, extending multi-layer readouts to NSD and EEG, and nonlinear readouts for higher-level areas.

References

[1] Allen, E. J. et al. (2022). A massive 7T fMRI dataset to bridge cognitive neuroscience and AI. *Nature Neuroscience*, 25, 116–126.

[2] van Bree, S. et al. (2025). Noise ceiling estimation for brain–model alignment. *Preprint*.

[3] Conwell, C. et al. (2024). Encoding RSA and CKA as hybrid alignment metrics. *Preprint*.